

Mark schemes

Q1.

(a) $(5\cosh x - 3\sinh x) = \frac{5}{2}(e^x + e^{-x}) - \frac{3}{2}(e^x - e^{-x})$
cosh x and sinh x correct in terms of e^x

M1

$= e^x + 4e^{-x}$
may be seen as denominator

A1

$\frac{1}{5\cosh x - 3\sinh x} = \frac{e^x}{4 + e^{2x}}$

*** must have left hand-side; $m = 4$*

A1cso

3

(b) $u = e^x \Rightarrow du = e^x dx$
 $\frac{du}{dx} = e^x$

M1

$\Rightarrow \int \frac{1}{4 + u^2} (du)$

FT "their" m from part(a) $\Rightarrow \int \frac{1}{m + u^2} du$

A1✓

$= \frac{1}{2} \tan^{-1} \frac{u}{2}$

FT "their" $\frac{1}{\sqrt{m}} \tan^{-1} \frac{u}{\sqrt{m}}$

A1✓

$x = 0 \Rightarrow u = 1 \quad x = \ln 2 \Rightarrow u = 2$

$\frac{1}{2} \tan^{-1} 1 - \frac{1}{2} \tan^{-1} \frac{1}{2}$

FT "their" $\frac{1}{\sqrt{m}} \left(\tan^{-1} \frac{2}{\sqrt{m}} - \tan^{-1} \frac{1}{\sqrt{m}} \right)$

A1✓

$= \frac{\pi}{8} - \frac{1}{2} \tan^{-1} \frac{1}{2}$



Q2.

$$(a) \quad \frac{dz}{dx} = 2 \cosh x \sinh x$$

Any correct method

M1

$$= \sinh 2x$$

AG

A1

2

$$(b) \quad I = \int_{x=0}^{x=1} \frac{dz}{1+z^2}$$

Ignore limits here

M1A1

$$= \left[\tan^{-1} z \right]_{x=0}^{x=1}$$

A1

$$= \left[\tan^{-1} (\cosh^2 x) \right]_0^1$$

Or A1 for change of limits

A1

$$= \tan^{-1}(\cosh^2 1) - \tan^{-1}(\cosh^2 0)$$

$$= \tan^{-1}(\cosh^2 1) - \frac{\pi}{4}$$

AG

A1

5

Q3.

$$u = x - 2$$

$$du = dx \quad \text{or} \quad \frac{du}{dx} = 1$$

clearly seen

B1

$$32 + 4x - x^2 = 36 - u^2$$

If $32 + 4x - x^2$ is written as $36 - (x - 2)^2$, give B2

B1

$$\int \frac{du}{\sqrt{36 - u^2}} = \sin^{-1} \frac{u}{6}$$

allow if dx is used instead of du

M1

limits -3 and 3

or substitute back to give $\sin^{-1} \frac{x-2}{6}$

A1

$$I = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$$

A1

[5]

Examiner reports

Q1.

- (a) Almost everyone managed to use the correct exponential forms for $\cosh x$ and $\sinh x$ and simplified their expression to the form on the right hand side of the identity. However, in order to score full marks, it was necessary to show the expression on the left hand side of the identity at least on the first line of working. The best students wrote an appropriate conclusion such as “therefore

$$\frac{1}{5\cosh x - 3\sinh x} = \frac{e^x}{4 + e^{2x}} \text{ ”. Practically every student showed that } m = 4.$$

- (b) Most students carried out the first stage of the substitution process, differentiating e^x , in order to transform the integral. The majority obtained an integral of the form

$$\int \frac{1}{4+u^2} du$$

although quite a few omitted the du and this was penalised. Others retained the limits for x on the integral with respect to u and this was penalised also. The vast majority did the integration well and obtained the printed answer. Students must realise that when they are asked to obtain a printed answer then they must

include vital steps such as writing the value of the integral as $\frac{1}{2} \tan^{-1} \frac{1}{2} \tan^{-1} \frac{1}{2}$ before writing down the printed answer.

Q2.

Part (a) was usually correctly done. However, responses to part (b) were mixed. It was

clear that not all candidates were familiar with $\int \frac{du}{1+u^2}$, even though it is given on page 8 of the formulae booklet. A significant number of candidates gave this integral as $\ln(1 + \cosh 4x)$.

Q3.

On the whole, responses to this question were poor. Many candidates were unable to handle the substitution for u in order to arrive at $36 - u^2$ and many sign errors occurred in

their working. Even of those candidates who arrived at $\int \frac{du}{\sqrt{36 - u^2}}$, many seemed

unable to connect the integral with algebraic forms appeared regularly. $\sin^{-1}\left(\frac{u}{6}\right)$ and $\sinh^{-1}\left(\frac{u}{6}\right)$ or $\cosh^{-1}\left(\frac{u}{6}\right)$ or even merely